## a Frozen-stock AND closure on original target populations

MCTS  Greedy I  Greedy II

Recorded-run lower bounds; unequal work; unfinished searches remain pending.

| URSA Expert | URSA Drugs | RetroCast | PaRoutes |
|---|---|---|---|
| 23/100 | 49/100 | 105/160 | 22/100 |
| 1/100 | 15/100 | 19/160 | 3/100 |
| 1/100 | 15/100 | 17/160 | 3/100 |
| MCTS pending: 77 | MCTS pending: 51 | MCTS pending: 54 | MCTS pending: 78 |

## b Experimental-reaction evidence along complete stock routes (targets)

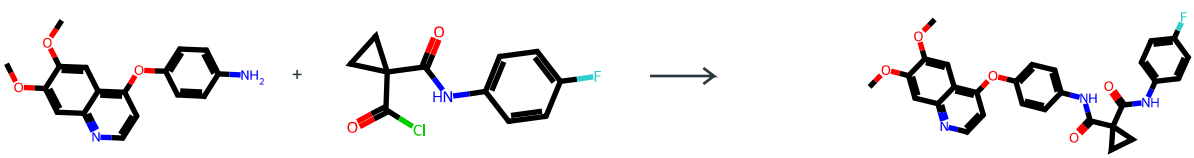
| Policy | Stock AND | Any exact step | Every step exact | Every step A/B |
|---|---|---|---|---|
| MCTS | 199 | 2 | 1 | 1 |
| Greedy I | 38 | 0 | 0 | 0 |
| Greedy II | 36 | 0 | 0 | 0 |

All-run reaction evidence: A=4, B=0, C=18206, D=0; n=18,210 distinct reactions.

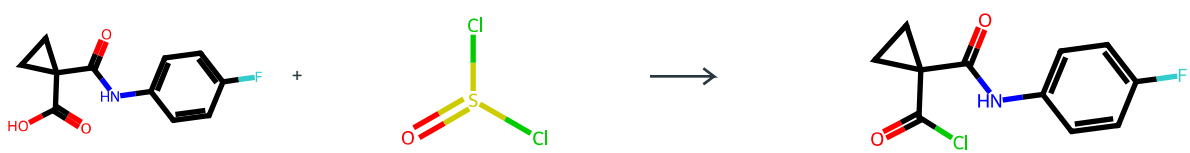
C denotes unresolved evidence, not chemical invalidity. Evidence is for whole reactions, not arrows.

## c Actual generated net transformations; source scope retained

1. Tier A · US20160022662A1



2. Tier A · WO2018104954A1



Net schemes: forward orientation. Symbolic inverse programs and conditions: linked case plates.